

To Shrink or to Select? Evaluating Finite-Sample Performance in Set-Identified Structural VARs

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Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

Keywords: Structural Vector Autoregression (SVAR), Set Identification, Bayesian VAR (BVAR), Lag Order Selection, Monte Carlo Simulation

JEL Codes: C11, C15, C32, C52

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1 Introduction

Structural Vector Autoregressions (VARs) remain the cornerstone of a large part of empirical macroeconomics, especially for monetary policy, shock transmission, and financial markets. While forecasting may be the main goal for some researchers, many economists also use it to estimate the relationships among factors that determine macroeconomic movements. For this purpose, Impulse Response Functions (IRFs) are derived from the VAR parameters, establishing how a shock in one variable affects the development of another period, both contemporaneously and over horizons of several months or even years. Using this transformation, structural VARs are continually adapted to answer complex questions, such as how speculative trading and real-time news shocks drive global oil market dynamics (Kilian and Murphy, 2014; Gazzani et al., 2024; Castelnovo et al., 2024), how high-frequency monetary policy surprises transmit through corporate credit spreads (Gertler and Karadi, 2015; Ferreira et al., 2023), and how these policy shifts impact global commodity prices (Miranda-Pinto, 2023). The specification of the VAR model, as well as the method to decompose it into its structural form, are the two crucial specifications that a researcher has to decide on. Traditionally, VAR models were specified with a lag order determined by economic theory or an information criterion (IC) and then estimated by Ordinary Least Squares (OLS) or Maximum Likelihood (ML). The past 15 years have seen a surge in the popularity of Bayesian methods, in which not a single lag order is specified, but lag orders are weighted and/or shrunk based on a prior and a posterior probability distribution (Blake and Mumtaz, 2015). These methods enable the researcher not only to rely on a single specification but also to incorporate a larger model or set of models while still keeping variance low. The methodology for estimating IRFs has also advanced significantly since Sims (1980) showed that structural IRFs can be retrieved via Cholesky decomposition, i.e., a predetermined causal ordering of the relationships between variables. Newer methods typically require weaker assumptions and adopt a more agnostic approach to inferring economic relationships. The dominant approach nowadays is set identification using sign restrictions informed by economic theory (Uhlig, 2005; Rubio-Ramírez et al., 2010). Unlike in Cholesky decomposition, all variables can simultaneously react to each other’s shocks, and sign restrictions can reduce the set of possible IRFs to those that fit the researcher’s assumptions. This yields a set of admissible IRFs. Typically, the median target—the specific IRF closest to the point-wise median of the admissible set—is selected as the primary result (Fry and Pagan, 2011).

This study draws inspiration from Ivanov and Kilian (2005), the first paper to examine which VAR specification best retrieves the true IRF using Monte Carlo simulations. In 2005, this meant testing criteria for lag selection to assess how well they fit VAR models and retrieve structural IRFs via Cholesky decomposition. We aim to update this analysis to the more recent world of VAR methodology, where IRFs are identified via

sign restrictions and estimation relies heavily on Bayesian methods rather than solely on information criteria.. We construct a realistic DGP based on Kilian and Murphy (2014)’s study of the oil market, using the set identification method. Then we compare different methods across repeated simulations by how well they can retrieve the true IRF, using the median target of their identified set. This study compares the information criteria AIC, BIC, and HQC with BVAR estimates using different overall shrinkage hyperparameters λ_1 . We find that BVARs are a well-performing alternative to OLS-estimated VAR models with IC-selected lag orders, beating the best criterion, AIC, in most scenarios. We could not deduce a clear pattern for which λ_1 -values shall be optimally employed depending on sample size T and true lag order p_0 . We attribute this to the fact that both higher and lower shrinkage can have the same overall effect: one reduces variance, the other reduces bias. Additionally, we suspect that our simulation still contains too much sampling variance, as computational restrictions did not allow for sufficient iterations. Optimizing λ_1 via the marginal data density (MDD), a metric geared toward out-of-sample forecasting, generally favors tighter λ_1 -values in order to shrink strongly. Taking the results for different λ_1 specifications into account, this is found to be suboptimal, although still an improvement over AIC in most cases. We hence show that the goals of forecasting and IRF analysis do not necessarily work together in VAR estimation.

The rest of the paper is structured as follows: Section 2 introduces the reader to the theoretical foundations needed to understand the methodology of this simulation study. It first introduces the VAR framework, Section 2.1, then explains the sign restriction approach for IRF decomposition, Section 2.2, and explains information criteria as well as the BVAR method, Section 2.3 and 2.4. We aim to both convey the intuition behind these methods as well as derive key properties. Section 3 explains the simulation design and Section 4 presents our results. Finally, the conclusion, Section 5, summarizes and discusses our key findings.

2 Theoretical Background

2.1 VAR Framework and IRFs

The reduced form of a vector autoregression, $\text{VAR}(p_0)$, establishes a relationship between current observations of a set of variables with the past observations of these variables up until a pre-determined lag p_0 :

$$Y_t = v + A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_{p_0} Y_{t-p_0} + e_t \quad (1)$$

Here, Y_t is a $(K \times 1)$ vector of endogenous variables, v is a vector of intercepts, $A_1, \dots, A_{p_0}$ are $(K \times K)$ coefficient matrices, e_t is a white-noise innovation vector with $\mathbb{E}[e_t] = 0$ and covariance matrix $\mathbb{E}[e_t e_t'] = \Sigma$, and p_0 denotes the lag order. This is the

standard reduced-form representation of a VAR process (Lütkepohl, 2005; Sims, 1980). However, reduced-form innovations are composed of structural shocks and are therefore uninformative about the effects of any individual shock. To identify the structural shocks of interest, restrictive assumptions must be imposed. The relationship between the two is given by:

$$e_t = B\epsilon_t \quad (2)$$

where ϵ_t denotes the vector of mutually uncorrelated structural shocks (that are normalized to unit variance i.e $\mathbb{E}[\epsilon_t\epsilon_t'] = I$) and B is a matrix satisfying $\Sigma = BB'$. Since there are infinitely many candidate matrices B (for any valid factor $\tilde{B}$ and any orthogonal matrix D , the product $\tilde{B}D$ also satisfies $\Sigma = (\tilde{B}D)(\tilde{B}D)'$), restrictive assumptions are required to single out a unique B . The literature offers several strategies for this purpose, including short-run (recursive) restrictions (Sims, 1980; Christiano et al., 1999), long-run restrictions (Blanchard and Quah, 1989), and sign restrictions (Faust, 1998; Uhlig, 2005; Rubio-Ramírez et al., 2010); see Kilian and Lütkepohl (2017) for a comprehensive treatment. Following Ivanov and Kilian (2005), we do not commit to a single scheme mentioned above but instead adopt, for each data-generating process, the identifying assumptions used in the original study on which it is based.

In general, interpreting the dynamics implied by VAR coefficients is difficult. Therefore, practitioners use Impulse Response Functions (IRF) to summarize the main dynamics that they want to explain. To obtain the coefficient set that constructs the impulse response function we express the VAR model in terms of an infinite VMA process. Using (1):

$$Y_t = [I - A_1L - A_2L^2 - \dots - A_{p_0}L^{p_0}]^{-1}v + [I - A_1L - A_2L^2 - \dots - A_{p_0}L^{p_0}]^{-1}e_t \quad (3)$$

Which is equivalent to:

$$Y_t = \mu + \sum_{i=0}^{\infty} \Phi_i e_{t-i} \quad \text{and} \quad \Phi_0 = I_K \quad (4)$$

where $\mu = [I - A_1 - \dots - A_{p_0}]^{-1}v$ is the unconditional mean vector and the moving average coefficient matrices are obtained recursively as $\Phi_i = \sum_{j=1}^i A_j \Phi_{i-j}$ where $A_j = 0$ for $j > p_0$ (see appendix for detailed derivation). Finally, in order to obtain the structural impulse response functions, we plug the expression given in (2) into (4):

$$Y_t = \mu + \sum_{i=0}^{\infty} \Phi_i B \epsilon_{t-i} = \mu + \sum_{i=0}^{\infty} \Theta_i \epsilon_{t-i} \quad (5)$$

Here, the matrices Θ_i are the structural impulse response matrices, and they contain the coefficients required to construct the impulse response function. To illustrate, the

equation for the first endogenous variable can be written as:

$$Y_{1t} = \mu_1 + \sum_{k=1}^K \theta_{1k,0} \epsilon_{kt} + \sum_{k=1}^K \theta_{1k,1} \epsilon_{kt-1} + \sum_{k=1}^K \theta_{1k,2} \epsilon_{kt-2} + \dots \quad (6)$$

where $\theta_{jk,i}$ denotes the response of variable j to structural shock k at horizon i , i.e., the (j,k) element of Θ_i .

The structural impulse responses in (6) depend on the model through two distinct channels. First, the moving average coefficients Φ_i are determined by the reduced-form autoregressive coefficients $A_1, \dots, A_{p_0}$, so the impulse responses depend on the lag order p_0 ; in practice p_0 is unknown and must be estimated, which is the subject of the lag-order selection criteria discussed in Section 3. Second, the mapping from reduced-form to structural form impulse responses requires the identification of B introduced in (2). Since there are infinitely many candidate matrices B (for any valid factor $\tilde{B}$ and any orthogonal matrix D , the product $\tilde{B}D$ also satisfies $\Sigma = (\tilde{B}D)(\tilde{B}D)'$), restrictive assumptions are required to single out a unique B . The literature offers several strategies for this purpose, including short-run (recursive) restrictions, often computed by Cholesky decomposition (Sims, 1980; Christiano et al., 1999), long-run restrictions (Blanchard and Quah, 1989), and sign restrictions (Faust, 1998; Canova and Nicoló, 2002; Uhlig, 2005; Rubio-Ramírez et al., 2010); see Kilian and Lütkepohl (2017) for a comprehensive treatment. We calibrate our data-generating process to the model of Kilian and Murphy (2014) and accordingly employ a sign-restriction algorithm to identify the impact matrix (though we omit the elasticity bounds for computational feasibility). A key feature of this approach is that sign restrictions deliver only partial identification: rather than a single impact matrix, they identify a *set* of candidate matrices consistent with the a priori restrictions implied by economic theory. This raises several questions about the identification strategy, which are the focus of the next subsection.

2.2 Sign Restrictions

Point identification strategies, such as the Cholesky decomposition, impose an assumption restrictive enough to single out a unique impact matrix; in the Cholesky case, this is a recursive ordering that determines the contemporaneous causal structure among the endogenous variables. Uhlig (2005), however, criticizes such strategies on the grounds that these restrictive assumptions may not reflect the underlying economic reality, but instead may impose the very conclusions the researcher expects to find.

Uhlig (2005) proposes an alternative he terms an "agnostic" identification procedure. The agnostic approach imposes only minimal restrictions—and, crucially, leaves the response of the variable of central interest entirely unrestricted—so that the answer to the research question is determined by the data rather than by the identifying assumptions.

The restrictions that are imposed are deliberately weak: they consist only of sign restrictions on a few responses that can be reasonably agreed upon across economists. This minimalism comes at a cost, however. Imposing only sign restrictions does not single out a unique impact matrix; instead, it identifies a set of candidate impact matrices consistent with the assigned signs. This set-identification (or partial identification) strategy is less restrictive, but it comes at the cost of ambiguity, since the object of interest is no longer a single value but a range. Moreover, as emphasized by Uhlig (2005), sign restrictions introduce a further choice that point-identifying schemes do not: the horizon over which the restrictions are imposed. In addition to specifying the sign of a response on impact, the practitioner must decide how many subsequent periods that sign is required to hold—the longer this horizon, the more candidate models are discarded and the narrower the identified set becomes.

This trade-off is described by Manski (2003) as the Law of Decreasing Credibility: the credibility of inference decreases as the strength of the maintained assumptions increases. Point identification strategies impose stronger assumptions and are therefore more informative—they reduce the identified object to a single point—but they are less credible, since their validity rests entirely on assumptions that may not hold. Partial identification strategies, by contrast, are less informative—they deliver a trimmed range rather than a single element—but more credible, precisely because they rely on weaker and more defensible assumptions.

Here, we follow Kilian and Murphy (2014) with slight differences due to computational feasibility.¹ First, we fit VAR(p_0) model to the data of Kilian and Murphy (2014) and obtain the reduced-form autoregressive coefficients ($\hat{A}$) and the variance-covariance matrix of the reduced-form errors ($\hat{\Sigma}$). We then factor this covariance matrix using Cholesky decomposition ($\hat{\Sigma} = PP'$).² Next, following Rubio-Ramírez et al. (2010), we generate random orthogonal matrices: we draw a matrix of independent standard normal entries, compute its QR decomposition, and normalize the sign of each column so that the diagonal of (R) is positive. This procedure yields orthogonal matrices (Q) (with $QQ' = I_K$) that are uniformly distributed with respect to the Haar measure, ensuring that the space of admissible rotations is sampled without bias. For each draw we form

¹Kilian and Murphy (2014) argue that agnostic sign restrictions alone are not sufficient to narrow the set of admissible models adequately, and they therefore impose additional quantitative restrictions; bounds on the implied oil supply and demand elasticities. Additionally, they set dynamic restrictions for the horizon of 12 months. We omit the elasticity restrictions and reduce the dynamic restrictions to 3 months for reasons of computational feasibility: in our finite samples the acceptance rate under the elasticity bounds is prohibitively low, so that obtaining a sufficient number of admissible models in each of the many Monte Carlo iterations would require a computationally unachievable number of draws.

²Kilian and Murphy (2014) uses eigendecomposition here. However, it doesn't make any difference. Any two factors satisfying $BB' = \tilde{B}\tilde{B}' = \hat{\Sigma}$ differ only by an orthogonal rotation ($B = \tilde{B}D$). Since our procedure applies the full set of Haar-distributed rotations to the initial factor, the resulting set of admissible models is invariant to the initial decomposition. Only element we need is an initial factor to run the algorithm.

a candidate impact matrix ($\tilde{B} = PQ$) and check whether it satisfies the imposed sign restrictions; candidates that violate the restrictions are discarded. Finally, we retain a select number of admissible impulse response structures and select the median target as our true IRF (population response). For the sake of clarity, we find it appropriate to outline the step-by-step blueprint of this algorithm:

1. Draw a $(K \times K)$ matrix W , where each element is drawn independently from a standard normal distribution: $W_{ij} \sim N(0, 1)$.
2. Compute the QR decomposition $W = QR$, where Q is an orthogonal matrix (the rotation matrix) and R is upper triangular. Normalize the sign of each column of Q so that the diagonal of R is positive, which renders the decomposition unique and ensures that Q is distributed according to the Haar measure.
3. Construct the candidate impact matrix $\tilde{B} = PQ$, and then check whether $\tilde{B}$ satisfies the imposed restrictions.
4. Repeat this process by drawing new W matrices (i.e., obtaining new rotation matrices Q) until the desired number of admissible models is reached.
5. Among the retained models, select the median target m^* as the representative (true) model.

The selection of a representative model is a much-debated issue in the literature, particularly the comparison between the pointwise median and the median target. Since we must obtain a single population response rather than a set—otherwise the mean squared error (MSE) could not be computed against a unique true IRF—we adopt the approach most appropriate for our design. Fry and Pagan (2011) criticize the pointwise median for a compelling reason: the pointwise median does not correspond to any single admissible model, as it combines responses drawn from different models with different impact matrices. Consequently, this synthetic profile may appear plausible but need not correspond to any B satisfying $\hat{\Sigma} = BB'$. This implies that the shocks may not be properly isolated, so the pointwise median cannot be used for variance decomposition. To address this, Fry and Pagan propose the median target approach:

$$m^* = \arg \min_m \sum_{j,k,i} \left(\frac{\theta_{jk,i}^{(m)} - \text{med}_{jk,i}}{\sigma_{jk,i}} \right)^2, \quad (7)$$

where $\text{med}_{jk,i} = \text{median}_m \{\theta_{jk,i}^{(m)}\}$, $\sigma_{jk,i} = \text{std}_m \{\theta_{jk,i}^{(m)}\}$.

Solving this minimization yields the admissible model whose impulse responses are closest to the pointwise median ($m^* \in M$). Since m^* is an actual admissible model rather than a synthetic construct, it satisfies $\hat{\Sigma} = BB'$ (which makes sure that shocks are orthogonalized) and can be used for variance decomposition. We adopt this approach due to the coherence concerns described above and take the median target as our representative

true IRF whereby its specific impact matrix $\tilde{B}$ determines the DGP.³

Identification is an essential part of recovering the underlying impulse response function. As noted earlier, however, it is not the only factor that matters: the specification of the reduced-form VAR also plays a role. Recall that $\Phi_i = \sum_{j=1}^i A_j \Phi_{i-j}$ where $A_j = 0$ for $j > p_0$, so the moving-average coefficients, and hence the impulse responses, depend on the lag order (p_0). The practitioner must therefore choose the lag order carefully. We address this in the next section.

2.3 Information Criteria

The true lag order p_0 is a priori unknown when estimating a reduced-form VAR; it is a matter of specification. Adding too many lags may lead to overfitting, which inflates the variance of the impulse response estimates by requiring the estimation of additional parameters. Underfitting is also problematic, as it introduces bias. The bias-variance trade-off is therefore central to lag-order selection. Traditionally, information criteria are used to address this issue, and their working principle is essentially the same (Lütkepohl, 2005, chap. 4):

$$\text{IC}(p) = \ln |\hat{\Sigma}(p)| + \underbrace{c_T \frac{K^2 p}{N}}_{\text{penalty term}} \quad (8)$$

Information criteria are formulas that embody the bias-variance trade-off described above. The $\ln |\hat{\Sigma}(p)|$ term is the log-determinant of the reduced-form error covariance matrix, estimated by quasi-maximum likelihood; this term penalizes underfitting, since a model with too few lags leaves $\hat{\Sigma}(p)$ large and thus raises $\text{IC}(p)$. The Penalty weight c_T depends on the specific criterion, and the penalty term increases with the lag order; the second component therefore penalizes overfitting. The lag order is estimated by:

$$\hat{p} = \arg \min_p \text{IC}(p), \quad 1 \leq p \leq \bar{p} \quad (9)$$

Following Ivanov and Kilian (2005), we set the maximum lag order to $\bar{p} = 12$ for our monthly data, corresponding to one year. This choice exceeds the largest true lag order considered $p_0 = 10$, ensuring that all criteria can both identify and overfit relative to the true order, and it makes our results directly comparable to those of Ivanov and Kilian (2005).⁴

³For a visualization of the IRFs selected for the DGP, see Figure 4 in the Appendix.

⁴We only include AIC, BIC and HQC as information criteria and omit the Likelihood Ratio tests and the Lagrange Multiplier test for lag selection as these have become practically irrelevant in the last 20 years for macroeconomic time series analysis and were also shown to perform poorly in Ivanov and Kilian (2005)

Although there are different types of $IC(p)$, three are mainly used:

$$AIC(p) = \ln |\hat{\Sigma}(p)| + \frac{2}{N} K^2 p \quad (10)$$

The Akaike Information Criterion (AIC) (Akaike, 1974) is the least overfitting-averse of these three criteria. It uses $c_T = 2$ and tends to overfit in practice.⁵

$$BIC(p) = \ln |\hat{\Sigma}(p)| + \frac{\ln N}{N} K^2 p \quad (11)$$

The Bayesian Information Criterion (BIC), also called the Schwarz Information Criterion (Schwarz, 1978), penalizes overfitting the most. It uses $c_T = \ln N$ and is a consistent estimator of the true lag order $\left(\text{plim}_{N \rightarrow \infty} \hat{p}^{\text{BIC}} = p_0\right)$.

$$HQC(p) = \ln |\hat{\Sigma}(p)| + \frac{2 \ln \ln N}{N} K^2 p \quad (12)$$

The Hannan-Quinn Information Criterion (Hannan and Quinn, 1979) is neither as harsh against overfitting as AIC nor as tolerant as BIC. It uses $c_T = 2 \ln \ln N$ as its penalty weight and, like BIC, is a consistent estimator of the true lag order.

The three criteria share the same fit term and differ only in the penalty weight c_T , which equals 2 for the AIC, $2 \ln \ln N$ for the HQC, and $\ln N$ for the BIC. A heavier penalty makes each additional lag more costly and therefore yields a more parsimonious model. Comparing the weights gives:

$$2 \leq 2 \ln \ln N \leq \ln N \quad (13)$$

where the first inequality holds for $N \geq 16$ and the second for all N .⁶ Consequently, the implied lag-order estimates satisfy:

$$\hat{p}^{\text{BIC}} \leq \hat{p}^{\text{HQC}} \leq \hat{p}^{\text{AIC}} \quad (14)$$

so that the BIC is the most parsimonious criterion and the AIC the least, with the HQC in between (Ivanov and Kilian, 2005). Since $N \geq 16$ holds across all sample sizes in our study, this ordering applies throughout. As seen in Table 1, our Monte Carlo Simulation confirms this ordering. The penalization of BIC is evidenced to be so strong

⁵In the literature, the AIC is generally regarded as prone to overfitting. In the VAR setting, however, this tendency is attenuated: adding one more lag requires estimating an entire $(K \times K)$ coefficient matrix, so the penalty grows by $\frac{2K^2}{N}$ at each step. Because each additional lag is this costly, even the AIC—the most lenient of the three criteria—can underfit in practice.

⁶The weights coincide at $N = e^e \approx 15.2$ (for $2 = 2 \ln \ln N$) and never cross for $\ln N$ versus $2 \ln \ln N$, since $x - 2 \ln x > 0$ for all $x > 0$. The condition $\ln N \geq 2$ (BIC versus AIC) holds for $N \geq 8$. Therefore, $N \geq 16$ is the requirement for (14) to hold.

that the lag order selected, $\hat{p}^{\text{BIC}}$, always stays close to 2, regardless of the sample size T and the true lag order p_0 . $\hat{p}^{\text{HQC}}$ lies above $\hat{p}^{\text{BIC}}$ and increases more apparently with T , however it also does not come close to converging to the true lag order. On the other hand, $\hat{p}^{\text{AIC}}$ is consistently much closer to the true lag order and has already nearly reached p_0 at $T = 600$. Therefore, although AIC is not asymptotically consistent, it is the only IC that can come close to the true lag order for realistic sample sizes.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

p_0	Estimator		AIC	BIC	HQC
	T				
4	240		2.99	1.95	2.06
	360		3.39	2.00	2.16
	480		3.61	2.01	2.40
	600		3.80	2.02	2.66
10	240		6.54	1.96	2.07
	360		8.53	2.00	2.23
	480		8.96	2.00	2.66
	600		9.16	2.02	4.03

Lütkepohl (1985) compares information criteria based on their ability to recover the true lag order. Ivanov and Kilian (2005), by contrast, object to this approach on the grounds of the bias-variance trade-off discussed above. To see why, it is helpful to consider the decomposition of the mean squared error (MSE) of the estimated median target IRF $\hat{\theta}$ compared to the true DGP IRF θ :

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = E[(\hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta)^2] \quad (15)$$

which can be reformulated to:

$$MSE(\hat{\theta}) = E \left[\left(\hat{\theta} - E[\hat{\theta}] \right)^2 + \underbrace{2 \left(\hat{\theta} - E[\hat{\theta}] \right) \left(E[\hat{\theta}] - \theta \right)}_{=0} + \left(E[\hat{\theta}] - \theta \right)^2 \right] \quad (16)$$

and finally, after we distribute the expected value operator:

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + Bias(\hat{\theta})^2 \quad (17)$$

we have derived an equation for the bias-variance-tradeoff. In small samples, if underfitting produces a reduction in variance large enough to offset the resulting increase in bias, the MSE falls; and if we take the MSE as a proxy for accuracy, this matters more than a criterion's ability to recover the true lag order. The lag order selected by a criterion that underfits may therefore outperform the true lag order; one of the central

remarks of Ivanov and Kilian (2005).

2.4 Bayesian VAR (BVAR)

3 Simulation Design

This Monte Carlo simulation study follows Ivanov and Kilian (2005) design with several alterations in order to make our extensions feasible and to fit the needs of our analysis. First of all, we only use one monthly DGP based on the empirical VAR study of Kilian and Murphy (2014). Therefore, we do not get estimations for quarterly VAR series or VEC series, and additionally, we do not have additional robustness by including several different series with different macroeconomic indicators and movements. The reason for this limitation is our tight time constraint. We use the dataset from Kilian and Murphy (2014) containing the following 4 variables: percent change in global oil production, real activity index from Kilian (2009), the log real price of oil, and changes in OECD crude oil inventories. Just as in the original study, we first estimate a VAR with a set lag order of these 4 variables and then use the sign restrictions approach explained earlier to identify structural IRFs and select the median target.

The DGP is formed by structural VAR model using the structural impact matrix $\tilde{B}$ that corresponds to the median target IRF as the parameter for the structural shocks e_t , which are drawn from an i.i.d. standard normal distribution. The formula for the DGP is represented by a structural VAR model using the estimated reduced form parameters A (for the lag coefficients y_{t-p}) and V (for the monthly dummy parameters d_t) and the structural impact matrix $\tilde{B}$ as follows:

$$y_t = Vd_t + Ay_{t-p} + \tilde{B}e_t \quad (18)$$

We estimate these parameters using the original data for two different lag orders, $p_0 \in \{4, 10\}$ and create DGPs with different sample sizes, $T \in \{240, 360, 480, 600\}$, where $T = N + \bar{p}$. $\bar{p}$ is the set of lag orders that are considered for both the IC and the BVAR estimations; it is uniformly set to $\bar{p} = 12$ to ensure that the true lag order p_0 is part of the set. We restrict the lag order selection p_0 to only include one high and one low lag order and drop $T = 300$ in comparison to the setup of Ivanov and Kilian (2005) in order to save computational power for the sign restriction decomposition. For sampling, we use random draws of length p_0 from the original dataset as initial values, then apply the parametric bootstrap using the SVAR described above and generate sequences of lengths T .

On each of these samples, VARs fitted by lag orders selected by both the three ICs as well as different specifications of BVAR are estimated, a fixed number of admissible IRFs found, and the median target IRF determined. We estimate all BVARs using $p = \bar{p} = 12$

as the lag order, thereby solely relying on shrinkage for reducing the variance and bias of the estimation. The selection of the overall shrinkage parameter λ_1 is the key criterion to determine how loose or how tight the prior is applied and to what extent further lag orders are shrunk towards 0. We apply $\lambda_1 \in \{0.2, 0.4, 0.6, 0.8\}$ to determine how optimal shrinkage depends on sample size T and true lag order p_0 . Additionally, we use the marginal data density (MDD) method to determine a λ_1 optimized on Out-of-Sample MSE for which we also provide the same set of admissible λ_1 -values. The limitation of the value-field is established for computational reasons; it captures a good range of typically used λ_1 -values, both tighter and looser specifications.

Finally, we estimate the MSE by calculating the geometric mean over the ratio of the squared errors of the median target IRFs $\hat{\theta}_j$ computed by method j in relation to the squared errors of the median target IRF by a model fit with the true lag order $\hat{\theta}_{p_0 i}$ over all Monte Carlo iterations from 1, ..., R :

$$MSE_j^{relative} = \exp \left(\frac{1}{n} \sum_{i=1}^R \ln \left(\frac{SE_j i}{SE_{p_0} i} \right) \right) \quad (19)$$

The geometric mean is preferred over the arithmetic mean as the ratios are bounded on the left side at 0, but unbounded on the right side. Outliers on the right side are hence overweighed in the arithmetic mean. Normalization by the model fit with p_0 ensures that performance in each iteration is weighed by how well a correctly identified model performs.

For the computation of the DGP parameters, we set the number of selected admissible IRFs to 1 million in order to get a highly accurate representation of the real world movements. However, during the Monte Carlo simulation we have to reduce that number to 1,000, which is seen as the minimum requirement for accurate set identification, albeit still with a larger variance. We set the number of Monte Carlo iterations to $R = 500$. We understand the concern that this may be too little for eliminating all Monte Carlo biases. Yet we have to rely on such a small number as it is the absolute maximum that our hardware allows for, even though we have optimized the code by employing parallel processing and delegation to C-based processes using the `numba` and `joblib` packages in `python`. All in all, this simulation still took 18 hours to complete with 6.5 million admissible IRFs drawn.

4 Results

First, we will look at the main comparison of performance between the different ICs and BVAR specifications in terms of recovering an accurate median target IRF. Table 2 displays the full set of relative MSEs. It reveals two main findings:

1. We confirm the results from Ivanov and Kilian (2005) about the performance of

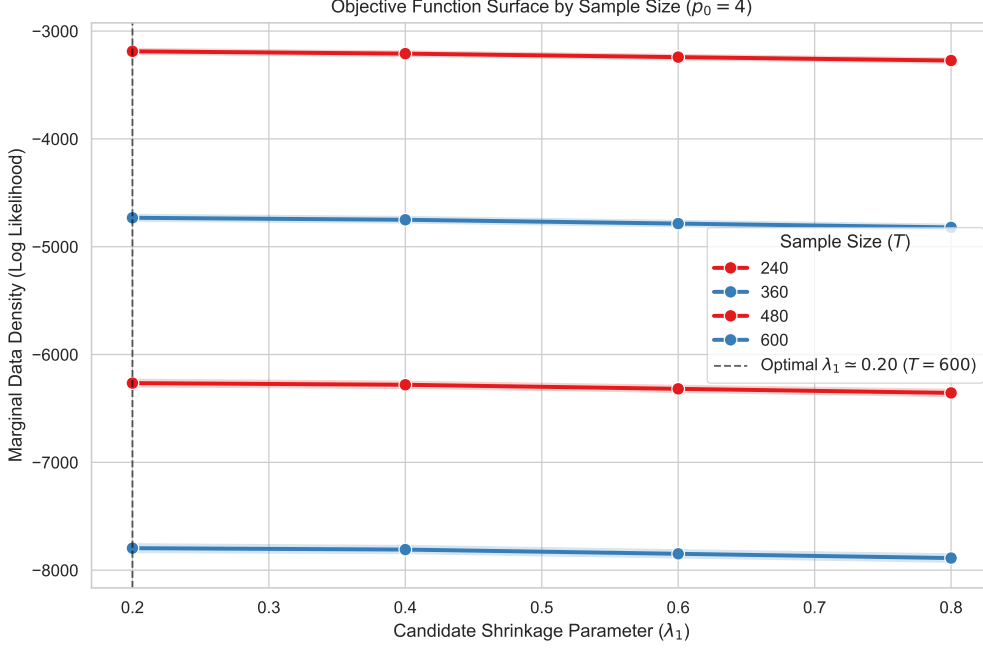


Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

the different information criteria. AIC consistently outperforms both BIC and HQC, revealing that the strict penalization of the latter two towards low lag orders is biasing the IRF too much. On the other hand, AIC's milder lag penalization ensures a closer fit to the true lag order p_0 for finite samples, thereby reducing bias. The reduction of the variance by selecting a p much lower than the true lag order selected by BIC and HQC, see Table 1, is not evidenced to reduce the coefficient variance enough to compensate for the additional bias.

2. BVAR dominates the three classical lag selection criteria across most scenarios; only in two cases AIC remains the leader amongst the methods as you can see in Table 2. For a better comparison amongst the different BVAR specifications, Table 3 presents the results in terms of relative MSE compared to the overall best performing information criterion, AIC. We determine that BVAR can outperform the AIC-fitted VAR by reducing the MSE by more than 20 percent in the most beneficial scenario with the best λ_1 -selection. On the other hand, a poor λ_1 choice may render the median target IRF approximately 15 percent less accurate than the AIC model. Overall, the BVAR approach with $\lambda_1 \in \{0.2, 0.4, 0.6\}$ has more instances of outperforming than underperforming AIC while $\lambda_1 = 0.8$ loses the comparison to the AIC benchmark in the majority of cases.

Overall the medium-loose prior $\lambda_1 = 0.6$ is the best hyperparameter in four cases. Both $\lambda_1 = 0.2$ and $\lambda_1 = 0.4$ are dominant in two scenarios. The very loose prior of $\lambda_1 = 0.8$ is shown to be optimal in no case. There is no absolutely clear pattern in terms of sample size or true lag order for why certain λ_1 -values perform better in certain

scenarios than in others.

This may be explained by two reasons. First, the inherent bias-variance trade-off in shrinkage implies that both strong shrinkage, through reduction of variance, as well as weak shrinkage, through reduction of bias, can make the estimation more precise. This may explain certain results in certain scenarios, such as the combination of p_0 and $T = 10$ where the relative MSE of the tightest hyperparameter, $\lambda_1 = 0.2$, is very close to the one of the loosest, $\lambda_1 = 0.8$. $\lambda_1 = 0.8$ ensures that higher lag orders are still more relevant for the estimation, making it fit better to the true model of lag order $p_0 = 10$. Meanwhile, $\lambda_1 = 0.2$ can reduce the variance of low lag orders, making a few coefficients more accurate while neglecting others. Second, the hardware-forced limitations on the number of iterations and number of required admissible IRFs per model may make the results too unstable. It can therefore not be refuted that some of the variation amongst our results are driven by random sampling variations.

Interestingly, the MDD-optimized λ_1 also never takes the first place amongst the BVAR models. Figure 1 reveals the reason. For low sample sizes T as well as the lower true $p_0 = 4$, MDD purely selects the lowest option $\lambda_1 = 0.2$ which has been shown to not perform the best in these cases. Only for $p_0 = 10$ does it select a higher $\lambda_1 = 0.4$ in some iterations. However, both $\lambda_1 = 0.2$ and $\lambda_1 = 0.4$ does not dominate for these scenarios. Again, the weak performance of these specific λ -values might be partially attested to sampling error. Nevertheless, it shows that MDD might not necessarily select the optimal hyperparameter for the purpose of the inference of structural relationships by VAR as its optimization on out-of-sample performance generally leads to a preference towards stronger shrinkage which might not be optimal for constructing IRFs as it eliminates the signal from higher lag orders.

Figure 2 illustrates how BVAR's MSE depends on the λ_1 in comparison to AIC for $T = 240$ and $T = 600$. While all shrinkage values ensure that BVAR outperforms AIC, the magnitude changes a lot. Table 3 shows the MSE of the different BVAR models relative to AIC's MSE. It

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Shrinkage.

T	p_0	AIC	BIC	HQC	BVAR				BVAR	
					$(\lambda_1 = 0.2)$	$(\lambda_1 = 0.4)$	$(\lambda_1 = 0.6)$	$(\lambda_1 = 0.8)$	$(\lambda_1 = 0.8)$	(MDD λ_1)
240	4	1.083	1.108	1.129	1.062	1.055	1.076	1.145	1.145	1.062
	10	1.033	1.018	1.021	0.895	0.887	0.868	0.947	0.947	0.895
360	4	1.117	1.274	1.296	1.030	1.167	0.994	1.142	1.142	1.030
	10	0.811	0.904	0.878	0.896	0.900	0.856	0.905	0.905	0.912
480	4	0.986	1.133	1.124	1.137	1.109	1.092	1.100	1.100	1.137
	10	0.989	1.190	1.185	0.853	0.920	0.876	0.944	0.944	0.866
600	4	0.989	1.145	1.118	1.133	0.983	1.103	1.109	1.109	1.133
	10	1.036	1.260	1.259	0.810	0.895	0.848	0.819	0.819	0.825

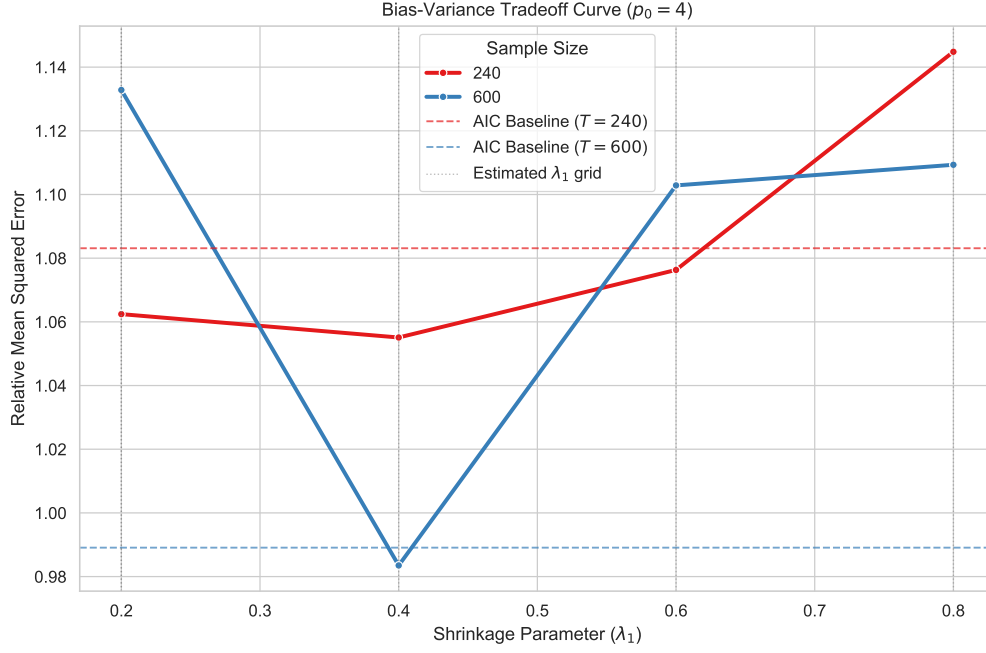


Figure 2: Comparison of Bias-Variance Tradeoff: BVAR vs. Hard Selection.

Table 3: Relative Mean Squared Error of Bayesian Shrinkage Specifications vs. AIC Baseline.

T	p_0	BVAR ($\lambda_1 = 0.20$)	BVAR ($\lambda_1 = 0.40$)	BVAR ($\lambda_1 = 0.60$)	BVAR ($\lambda_1 = 0.80$)	BVAR (MDD λ_1)
240	4	0.981	0.974	0.994	1.057	0.981
	10	0.866	0.858	0.840	0.917	0.866
360	4	0.922	1.045	0.890	1.022	0.922
	10	1.105	1.110	1.055	1.116	1.124
480	4	1.153	1.125	1.108	1.116	1.153
	10	0.863	0.931	0.886	0.955	0.876
600	4	1.145	0.994	1.115	1.122	1.145
	10	0.782	0.865	0.819	0.791	0.796

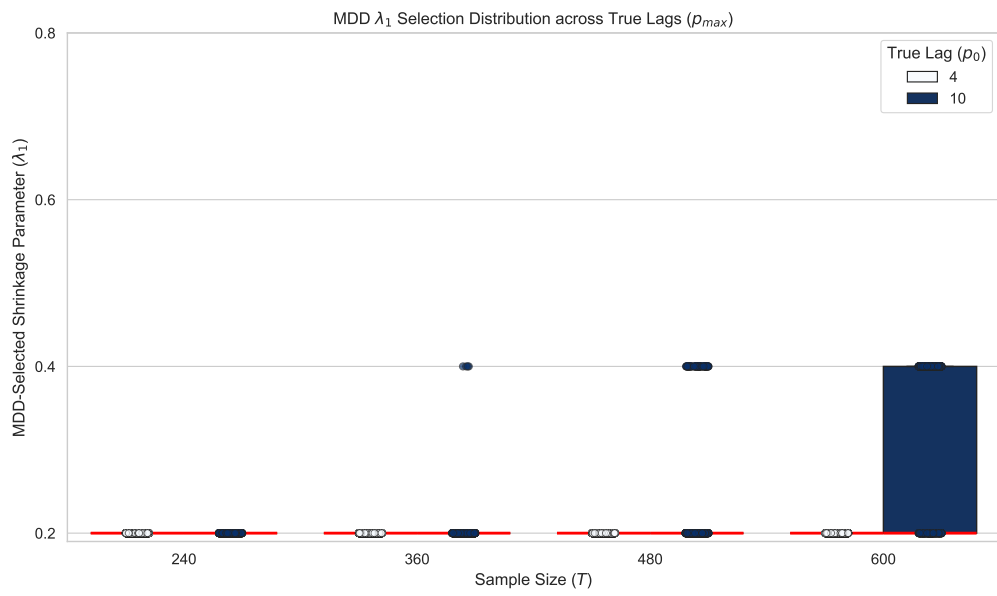


Figure 3: The MDD selected priors for different sample sizes and both $p_0 = 4$ and $p_0 = 10$.

5 Conclusion

Overall, we could show that BVARs can stand the test against traditional IC-fitted VAR models within a frequentist Monte Carlo setup. BVARs are shown to outperform even the best-performing information criterion, AIC, in most scenarios in retrieving a median target IRF close to the DGP IRF. We can also observe that the accuracy of the BVAR estimates depend strongly on the hyperparameter λ_1 , which defines the magnitude of parameter shrinkage. Yet, we cannot give concrete advice for researchers about which λ_1 -value to apply as our results are too instable. We attribute this to both the bias-variance-tradeoff of higher/lower shrinkage as well as to computational limitations on the number of iterations and IRF draws. Running Monte Carlo simulations with the goal of analyzing set identified IRFs remains a challenge due to the explosion of estimations by having to draw thousands of random B -matrices within each iteration.

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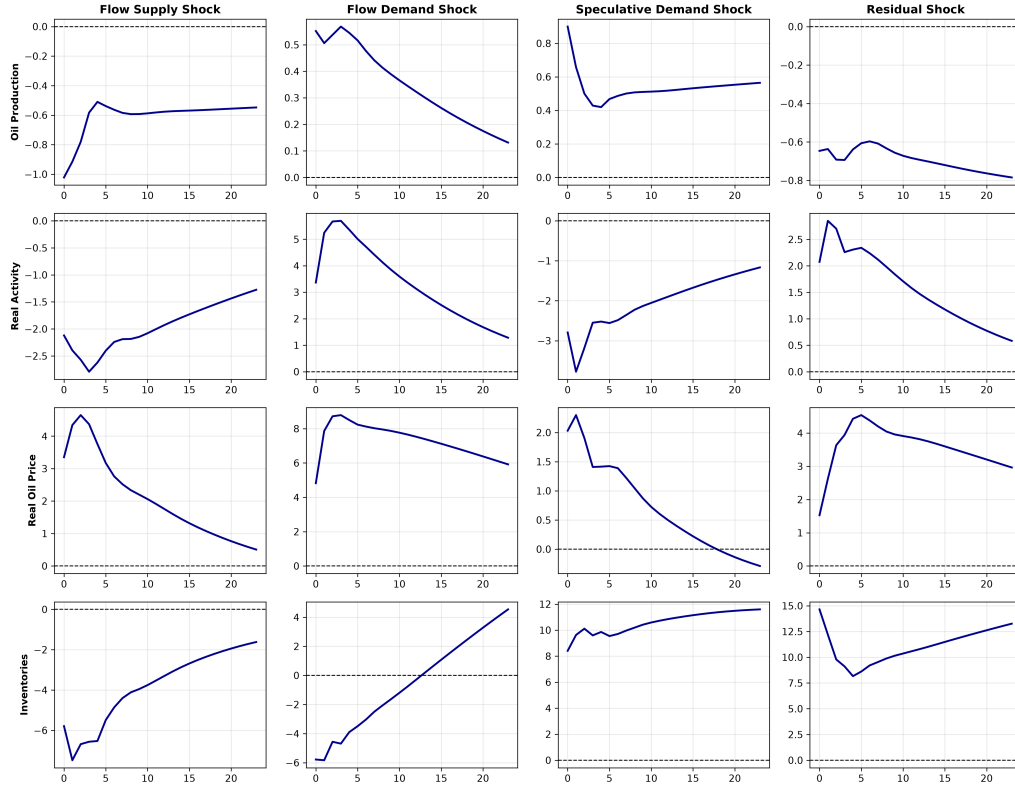
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Appendix

A. DGP IRFs

Figure 4 visualizes the IRFs selected for the DGP of $p_0 = 4$ and $p_0 = 10$. You can see that they exhibit the same general pattern but that a higher lag order captures more complex movements.

True Structural Impulse Responses (4-Lag Model, Median Target Model)



True Structural Impulse Responses (10-Lag Model, Median Target Model)

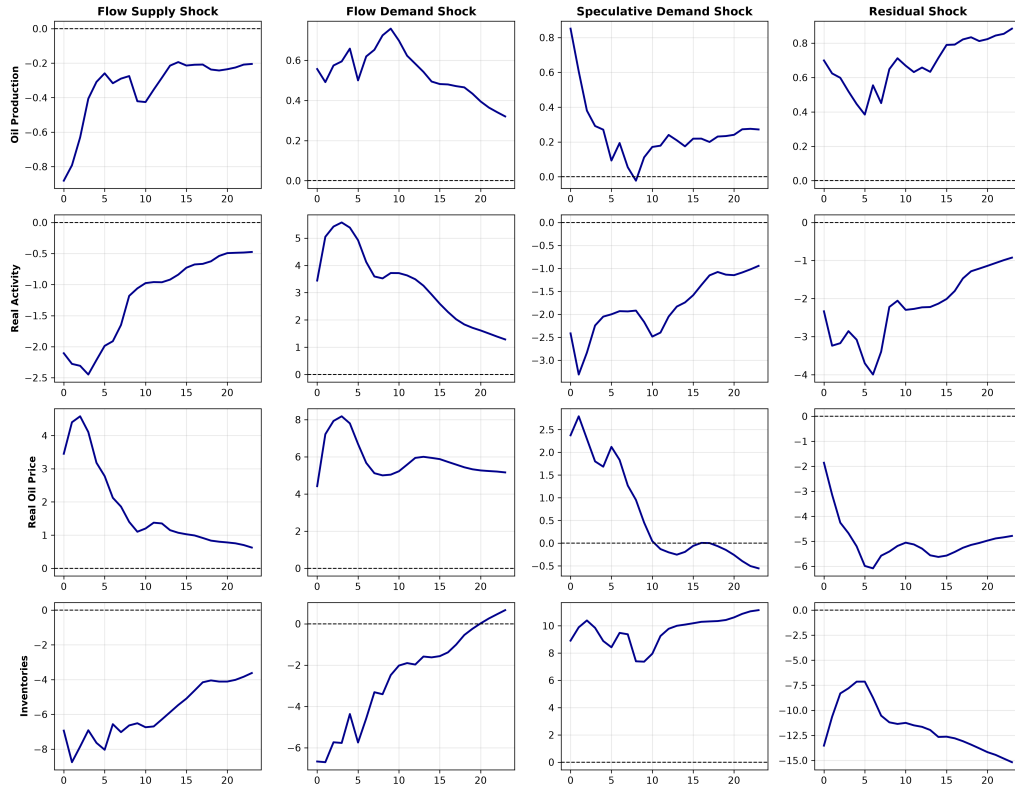


Figure 4: DGP IRFs for $p_0 = 4$ and $p_0 = 10$

B. Derivation of the Relationship Between Structural Form Innovation Terms and Reduced Form Errors

For clarity, several intermediate steps were omitted in the main text. We collect them here. To derive the relationship in equation (2), we begin from the structural form:

$$Y_t = B_0 Y_t + B_1 Y_{t-1} + B_2 Y_{t-2} + \cdots + B_{p_0} Y_{t-p_0} + \varepsilon_t \quad (\text{x})$$

Moving the contemporaneous term $B_0 Y_t$ to the left-hand side gives:

$$\underbrace{(I - B_0)}_{B^*} Y_t = B_1 Y_{t-1} + B_2 Y_{t-2} + \cdots + B_{p_0} Y_{t-p_0} + \varepsilon_t \quad (\text{x})$$

Premultiplying by the inverse of B^* yields the reduced form:

$$Y_t = \underbrace{(B^*)^{-1} B_1}_{A_1} Y_{t-1} + \underbrace{(B^*)^{-1} B_2}_{A_2} Y_{t-2} + \cdots + \underbrace{(B^*)^{-1} B_{p_0}}_{A_{p_0}} Y_{t-p_0} + \underbrace{(B^*)^{-1} \varepsilon_t}_{e_t} \quad (\text{x})$$

which is the explicit reduced-form representation. The relationship defined in (2) is now apparent: writing $(B^*)^{-1} = B$, we have:

$$e_t = B \varepsilon_t \quad (\text{x})$$

Here, B_0 and B denote distinct objects: B_0 is the structural contemporaneous coefficient matrix, which governs the simultaneous interactions among the endogenous variables in the structural form, whereas $B = (I - B_0)^{-1}$ is the impact matrix that maps the unit-variance structural shocks ε_t into the reduced-form innovations e_t . The former is a primitive of the structural model; the latter is the object recovered through the identification procedure in the main text.

C. Derivation of the Recursive Coefficients and the Mean

We now derive the recursion in (4). It is convenient to define the autoregressive lag polynomial:

$$A(L) = I - A_1 L - A_2 L^2 - \cdots - A_{p_0} L^{p_0} \quad (\text{x})$$

Its inverse admits a power-series representation in the lag operator:

$$A(L)^{-1} = \Phi(L) = \sum_{i=0}^{\infty} \Phi_i L^i = \Phi_0 + \Phi_1 L + \Phi_2 L^2 + \cdots \quad (\text{x})$$

Which leads to:

$$A(L)A(L)^{-1} = A(L)\Phi(L) = (I - A_1L - A_2L^2 - \dots)(\Phi_0 + \Phi_1L + \Phi_2L^2 + \dots) = I \quad (\text{x})$$

Matching coefficients on equal powers of L gives:

$$\begin{aligned} I\Phi_0 &= I_K \implies \Phi_0 = I_K \\ \Phi_1 - A_1\Phi_0 &= 0 \implies \Phi_1 = A_1\Phi_0 \\ \Phi_2 - A_1\Phi_1 - A_2\Phi_0 &= 0 \implies \Phi_2 = A_1\Phi_1 + A_2\Phi_0 \end{aligned}$$

and, in general, the recursion:

$$\Phi_i = \sum_{j=1}^i A_j \Phi_{i-j} \quad (\text{x})$$

The unconditional mean is $\mu = (I - A_1 - A_2 - \dots - A_{p_0})^{-1}v$ provided the process is weakly stationary. To derive it, take expectations of (1) under weak stationarity, so that $\mathbb{E}[Y_t] = \mu$ for all t :

$$\mathbb{E}[Y_t] = v + A_1\mathbb{E}[Y_{t-1}] + A_2\mathbb{E}[Y_{t-2}] + \dots + A_{p_0}\mathbb{E}[Y_{t-p_0}] \iff [I - A_1 - A_2 - \dots - A_{p_0}]\mu = v \quad (\text{x})$$

which gives:

$$\mu = [I - A_1 - A_2 - \dots - A_{p_0}]^{-1}v \quad (\text{x})$$

Declaration of AI Usage

During the preparation of this seminar paper, I, [NAME], used Claude, Gemini, and ChatGPT for help with coding, research, and adapting writing to academic standards. I have reviewed and edited the content as needed and take full responsibility for the content.

[NAME]